

STATESVILLE, NC, USA

WMO: 723055

Lat: 35.765N

Lon: 80.957W

Elev: 294

StdP: 97.84

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.5	(c) -5.5	(d) -18.7	(e) 0.7	(f) -2.3	(g) -16.0	(h) 1.0	(i) -1.4	(j) 8.3	(k) 6.2	(l) 7.5	(m) 6.2	(n) 1.3	(o) 300	(p) 0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	33.3	23.3	32.4	23.1	31.2	22.7	25.5	29.5	24.8	29.2	24.1	28.6	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.3	26.8	23.7	19.2	26.4	22.8	18.2	26.0	80.2	29.9	76.8	29.1	74.0	28.6	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 5.7	(c) 5.0	(e) -11.7	(f) 35.0	(g) 2.7	(h) 1.8	(i) -13.6	(j) 36.2	(k) -15.2	(l) 37.3	(m) -16.7	(n) 38.3	(o) -18.7	(p) 39.5		
(a) 7.2	(b) 5.7	(c) 5.0	(e) -12.6	(f) 26.4	(g) 2.7	(h) 1.1	(i) -14.5	(j) 27.1	(k) -16.1	(l) 27.7	(m) -17.6	(n) 28.3	(o) -19.5	(p) 29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	3.9	5.9	10.2	15.2	19.5	23.6	25.1	24.5	21.5	15.3	9.6	6.0
	DBStd	8.51	5.37	5.03	5.14	4.47	3.60	2.49	2.14	2.44	3.27	4.41	4.53	4.72
	HDD10.0	617	201	133	62	9	0	0	0	0	0	7	62	143
	HDD18.3	1965	448	348	254	112	32	1	0	0	10	114	264	382
	CDD10.0	2471	11	19	69	166	295	409	467	451	344	171	48	20
	CDD18.3	776	0	1	3	18	69	161	209	192	104	19	1	1
	CDH23.3	6373	1	2	29	189	561	1366	1757	1532	758	167	11	1
	CDH26.7	2304	0	0	2	33	146	518	720	596	254	36	0	0
Wind	WSAvg	1.7	2.0	2.1	2.3	2.3	1.7	1.5	1.3	1.2	1.5	1.6	1.7	1.7
Precipitation	PrecAvg	1154	92	91	113	87	98	112	100	100	97	86	81	93
	PrecMax	1556	236	180	221	222	213	260	197	273	278	321	161	188
	PrecMin	693	20	16	32	9	22	20	20	15	1	0	11	9
	PrecStd	172	45	40	50	48	52	55	49	51	66	64	39	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	23.0	26.3	29.4	31.9	34.6	34.7	36.0	33.6	30.3	24.7	22.3
		MCWB	14.6	16.1	15.6	17.5	21.0	23.1	23.4	23.9	22.0	21.8	18.6	16.2
	2%	DB	17.6	20.0	23.7	27.4	29.9	32.8	33.0	32.9	31.7	27.2	22.2	19.0
		MCWB	13.0	13.2	15.3	17.6	20.7	22.7	23.7	23.7	21.7	19.6	15.1	16.6
	5%	DB	15.4	17.6	21.8	25.8	28.2	31.4	32.2	31.9	29.8	25.1	20.2	16.5
		MCWB	11.0	12.3	14.5	17.0	20.0	22.3	23.6	23.1	21.6	18.7	15.5	12.9
	10%	DB	12.7	15.0	19.0	23.6	26.8	29.8	31.0	30.1	27.6	22.8	17.9	13.9
		MCWB	9.1	9.8	12.9	15.9	19.3	22.0	23.1	23.1	21.0	17.9	13.8	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	18.0	18.3	21.8	23.0	25.1	28.7	27.5	24.5	23.3	20.2	19.0
		MCDB	19.3	20.7	22.7	26.0	27.8	29.8	31.9	30.0	29.0	26.6	22.6	19.8
	2%	WB	14.4	15.8	17.0	20.0	22.0	24.1	25.7	25.6	23.6	22.0	18.2	16.6
		MCDB	16.4	18.6	21.3	24.6	27.5	29.4	29.4	29.6	28.1	25.3	20.1	19.0
	5%	WB	12.2	13.6	15.6	18.7	21.3	23.6	24.9	24.9	22.9	20.6	16.6	14.2
		MCDB	14.2	16.7	19.8	22.9	26.2	28.8	29.3	29.0	27.1	23.6	18.8	15.9
	10%	WB	9.6	10.5	14.4	17.3	20.4	22.9	24.1	24.1	22.2	19.0	14.4	11.4
		MCDB	12.5	13.7	18.0	21.6	24.7	27.9	28.7	28.1	26.0	21.7	17.3	13.0
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.1	11.7	12.4	11.6	11.1	10.5	10.1	10.3	12.0	12.4	10.3
		MCDBR	13.7	14.4	14.0	14.5	13.5	13.3	12.2	12.3	13.1	13.8	14.9	12.6
		MCWBR	9.5	9.5	7.6	7.1	6.0	4.7	4.5	4.3	4.8	7.0	9.4	8.9
	5% WB	MCDBR	11.0	12.3	12.0	12.1	11.3	11.3	10.7	10.3	10.4	11.1	11.5	10.3
MCWBR		9.2	10.0	8.1	7.0	5.5	4.9	5.8	5.4	5.1	6.8	9.5	8.4	
Clear-Sky Solar Irradiance	taub	0.308	0.319	0.353	0.400	0.451	0.481	0.530	0.519	0.419	0.372	0.323	0.308	
	taud	2.499	2.463	2.405	2.276	2.161	2.118	2.007	2.016	2.286	2.399	2.496	2.530	
	Ebn at Noon	895	924	916	884	839	812	769	768	836	850	868	872	
	Edh at Noon	82	95	110	132	151	157	175	170	123	101	82	75	
All-Sky Solar Radiation	RadAvg	2.56	3.21	4.27	5.33	5.92	6.32	5.89	5.42	4.58	3.75	2.84	2.19	
	RadStd	0.16	0.38	0.29	0.42	0.48	0.45	0.40	0.26	0.41	0.47	0.24	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+95

Nomenclature: See separate page